

Prebuilt

Precision printing for labs that can't share their data

Printers learn their errors from each other. Patient scans, part geometry and measurements never leave the lab.

ScienceClaw · Manufacturing & Materials · MIT Media Lab

THE PROBLEM

Labs that print can't share what they learn

Guides, crowns and implants from patient scans

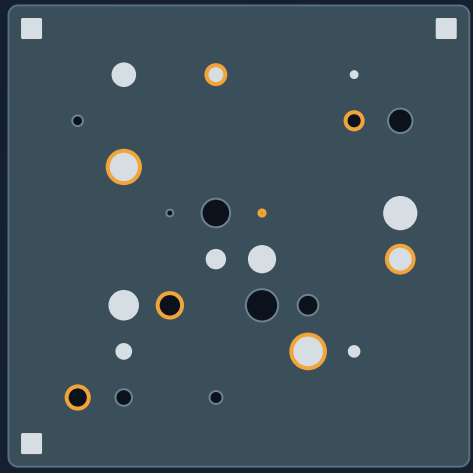
They must seat on the first print, and the scans are protected health data.

Microfluidic chips and bioprinted scaffolds

Sub-millimetre features, and every test print costs days and material.

Every printer is wrong in its own way

Holes print small and pins fat, by tens to hundreds of µm. Only printing reveals it.



The test plate: 24 pins and holes, 1–7 mm. Print it, measure it, know your printer.

Can printers learn from each other without sharing a single measurement?

HOW IT WORKS

Learn locally, share three numbers

1 Print a test plate

24 pins and holes. Measure them by micrometer or flatbed scanner.

2 Learn your printer

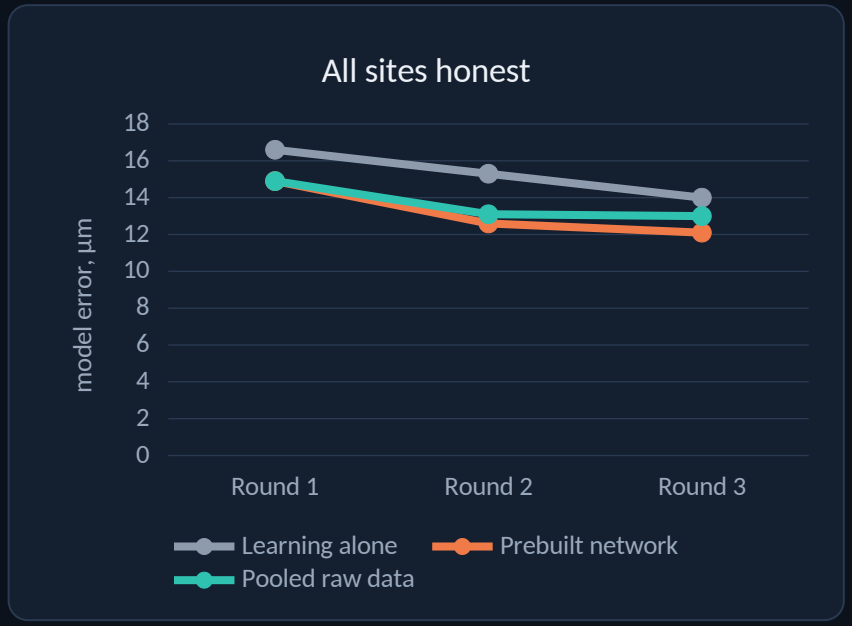
A physics model of your machine, fit locally on your own data.

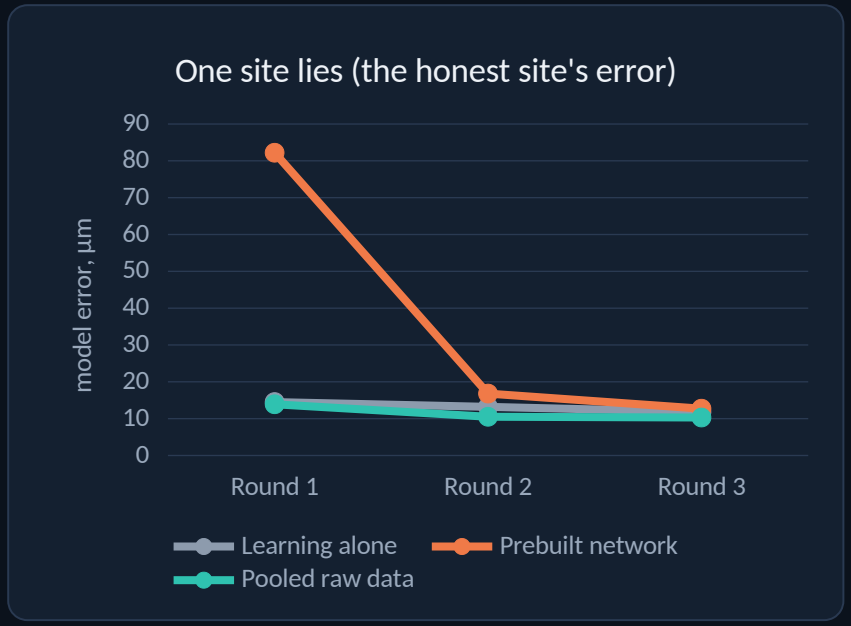
3 Share three numbers

Mean and spread of 3 parameters every printer shares. Schema-checked, logged on a receipt.

Never leaves the lab: patient scans, part geometry, measurements, machine settings. A replayable privacy audit checks every crossing: 0 measurements crossed.

The network learns faster, and shrugs off a liar





Average over 20 simulated fleets, registered config, error on held-out pieces. A liar spikes the honest site to 82.2 µm in round 1; trust collapses and it is back to 12.7 µm by round 3. With the consistency check on, round 1 stays at 13.3 µm.

RESULTS · SIMULATION

The win is many sites, not many agents

12%

lower prediction error than a lab learning alone. The network won in 75 of 100 simulated fleets. Pooling all raw data: also 12%.

35%

lower error with eight data-poor labs. The network won in 20 of 20 simulated fleets.

0 rounds

lost to a liar with the consistency check: 180 of 180 lies caught, 0 honest notes dropped.

9 µm

size error from a corrected design, against 132 µm drawn as-is (median over 60 simulated fleets).

Error: how far the model's predicted size of a pin or hole is from its true printed size, on held-out pieces it never trained on. A simulated fleet: printers whose true behavior is hidden from the model, run both ways and compared head to head. The agent layer adds nothing measurable.

A real printer for scale, simulation to score

Registered first

Claims and the prediction that two printers can't show the gain were committed before printing.

Negative control

When printers share no physics, sharing gains nothing. The gain isn't built into the scoring.

Why simulation

The two-site gain is 1.6 µm; two readings of one pin differ by 65 µm. Only hidden truth can score it.

REAL PRINTER · FORM STUDIO 1

1.44 mm

a 1.50 mm pin, by micrometer (two readings: 1.41 and 1.48)

- Inside the vendor's ±0.08 mm spec
- Inside the simulator, at its small edge
- One pin: a sanity check, not a model fit

WHO IT'S FOR

Medical and bio labs that print and can't pool data

Hospital 3D-printing labs

Surgical guides and anatomical models from patient scans. The scans stay in-house.

Dental labs

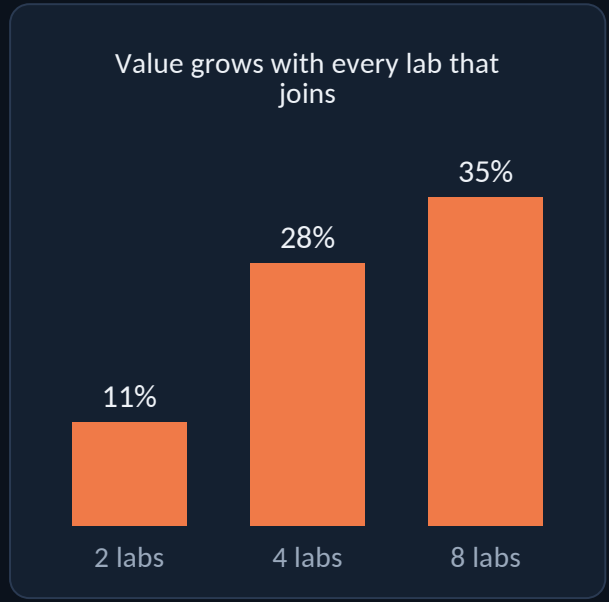
Crowns, models and aligners must seat first time. Patient data can't be pooled.

Bio and microfluidics labs

Lab-on-chip and organ-on-chip parts with sub-mm features. Recipes are IP.

Medical device makers

One part, many contract printers. Receipts double as an audit trail.



Buyers are our hypothesis, not yet validated with customers. Chart: lower error than learning alone at 2 / 4 / 8 data-poor sites, simulation. On sub-millimetre features the gain holds at 7–9% in simulation; bioprinting is direction, not a result.

VALIDATION AND BENCHMARKS

Many sites beat one; agents don't beat random

Round 3, simulated fleets, head to head	Lower error	Fleets where it wins	p
Network vs single site alone	12.3% (1.6 µm)	75 / 100	<0.001
Pooled raw data (upper bound) vs alone	12.2%	64 / 100	
Agents only, messages cut, vs alone	2.4%	56 / 100	0.27
Agent auction vs random pieces	+1.0%	31 / 60	0.90
4 / 8 data-poor sites vs alone	28.5% / 34.7%	35/40 · 20/20	<0.001
Negative control: nothing shared	−13.0% (adaptive spread −0.4%)	19 / 60	0.006

Hidden-truth simulator, registered config, error against true size on held-out pieces, sign test over worlds. 90% intervals cover 90–91%. GPU posterior up to 84× NumPy on an H100. Reproduce: docs/validation-and-benchmarks.md.



Print anywhere, trust the part

Labs learn their printers together. Their data stays home.

github.com/matte-realize/prebuilt · all sharing results in simulation